SHAPE AREA & PERIMETER CALCULATOR

OOPS Mini Project Report

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Introduction: The Shape Area and Perimeter Calculator is a simple C++ project that calculates the area and perimeter of different geometric shapes. The program supports three shapes: Circle, Rectangle, and Triangle.

The project uses Object-Oriented Programming concepts such as inheritance, virtual functions, function overriding, and runtime polymorphism. A common base class named Shape is created, and three shapes are derived from it.

The user selects a shape, enters its dimensions, and receives the calculated area and perimeter. The program also checks dimensions and rejects invalid triangle sides.

Objective: The main objective is to develop a C++ calculator that calculates the area and perimeter of different shapes through a common interface.

The specific objectives are:

1. To create a base class named Shape.
2. To define virtual functions for area and perimeter.
3. To create derived classes for Circle, Rectangle, and Triangle.
4. To demonstrate inheritance and function overriding.
5. To use a base-class pointer to demonstrate runtime polymorphism.
6. To accept dimensions from the user and display the results.
7. To handle invalid input and dimensions appropriately.

Problem Statement: Different geometric shapes have different formulas for calculating their area and perimeter. The purpose of this project is to develop one program that can perform these calculations for multiple shapes using a common interface.

The program accepts the dimensions of a circle, rectangle, or triangle. It then calculates and displays the corresponding area and perimeter. The project uses inheritance and runtime polymorphism to call the correct calculation functions for the selected shape.

Theoretical Background:

1. **Circle:** A circle is defined by its radius.

   $$\text{Area} = \pi \times r \times r$$

   $$\text{Perimeter} = 2 \times \pi \times r$$

   Here, r represents the radius and π is approximately 3.14159.

2. **Rectangle:** A rectangle is defined by its length and width.

   $$\text{Area} = \text{length} \times \text{width}$$

   $$\text{Perimeter} = 2 \times (\text{length} + \text{width})$$

3. **Triangle:** A triangle is defined by its three side lengths.

   $$\text{Perimeter} = a + b + c$$

   The area is calculated using Heron's formula:

   $$s = (a+b+c)/2$$

   $$\text{Area} = \sqrt{[s \times (s-a) \times (s-b) \times (s-c)]}$$

   Here, a, b, and c are the side lengths, and s is the semi-perimeter.

   For a valid triangle, the sum of any two sides must be greater than the third side.

OOPS Concepts Used:

1. **Class and Objects:** A class is a blueprint for creating objects. The Shape class provides a common interface, while Circle, Rectangle, and Triangle represent individual shapes.

   Objects of these derived classes store the dimensions needed for their calculations.

2. **Inheritance:** Inheritance allows a derived class to use the features of a base class. Circle, Rectangle, and Triangle inherit from Shape.

   This creates a common structure for calculating area and perimeter.

3. **Virtual functions:** The Shape class declare area() and perimeter() as virtual functions. These functions allow the correct derived-class implementation to be called through a base-class pointer.
4. **Function Overriding:** Each derived class provides its own implementation of area() and perimeter(). Circle uses the radius, Rectangle uses length and width, and Triangle uses three sides.
5. **Runtime Polymorphism:** Runtime polymorphism allows the same function call to produce different results depending on the actual object.

   In this project, a Shape pointer refers to a Circle, Rectangle, or Triangle object. Calling area() or perimeter() through this pointer invokes the appropriate overridden function.

6. **Encapsulation:** The dimensions of each derived class are declared private. This keeps the internal data protected from direct access outside the class.

Program Design:

The program consists of one base class and three derived classes

Shape: Declares the virtual functions area() and perimeter().

Rectangle: Stores the length and width and calculates its area and perimeter.

Triangle: Stores three sides and calculates the area using Heron's formula and the perimeter by adding the sides.

Main Function: Displays the menu, accepts user input, creates the required object, and displays the results through a Shape pointer.

A helper function checks that numeric dimensions are positive. The triangle calculation also checks the inequality before calculating its area.

Test Cases:

Test	Input	Expected result
1	Circle, radius=7	Area=153.94; perimeter=43.98
2	Rectangle, length=5, width=3	Area=15.00; perimeter=16.00
3	Triangle, sides=3,4,5	Area=6.00; perimeter=12.00
4	Triangle, sides=1,2,5	Invalid triangle message
5	Radius=0 or a negative value	Positive-number validation message
6	Menu input=abc	Invalid-choice message

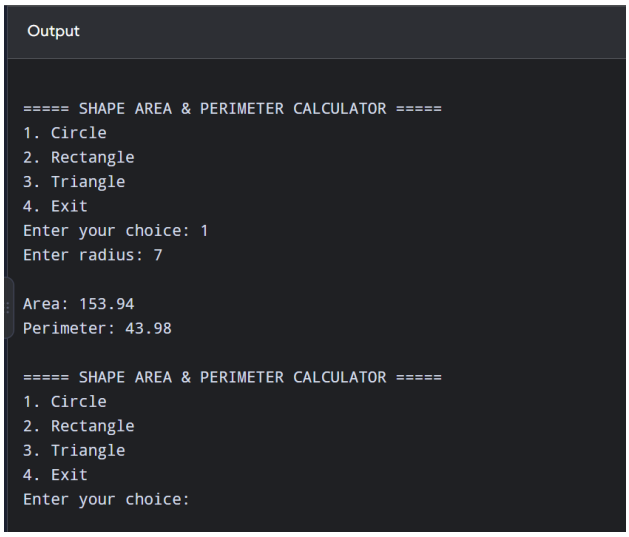
Conclusion:

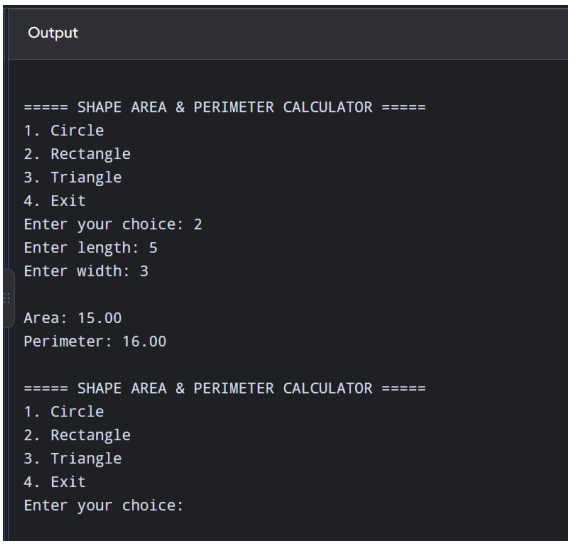
The Shape Area and Perimeter Calculator was developed using C++. It calculates the area and perimeter of circles, rectangles, and triangles through a common interface.

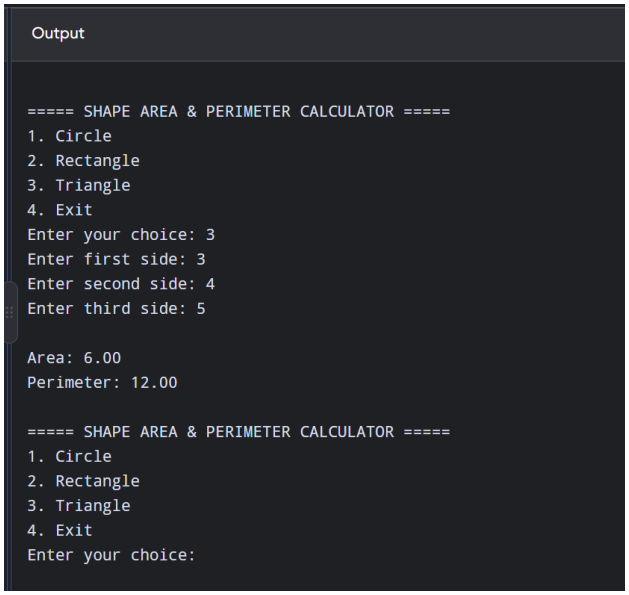
The project demonstrates inheritance, virtual functions, function overriding, runtime polymorphism, classes, objects, and encapsulation. It also uses input validation to handle invalid dimensions.

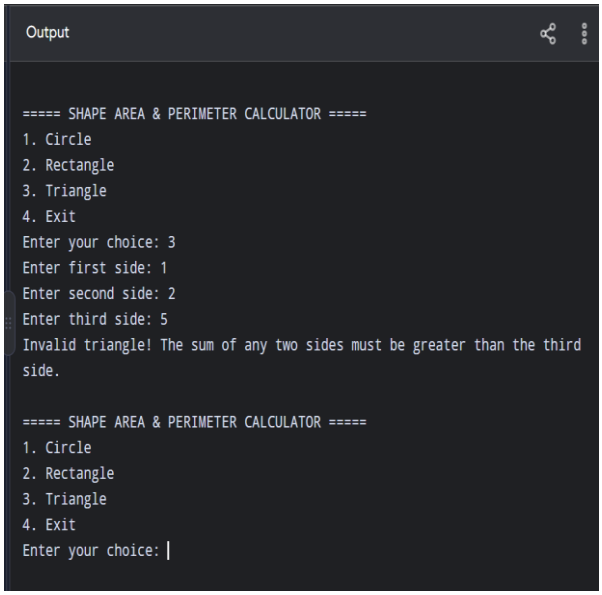
Overall, this project provides a practical understanding of how Object-Oriented Programming can be used to build a simple and reusable calculator.

OUTPUTS:


Output
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 1
Enter radius: 7
Area: 153.94
Perimeter: 43.98
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice:


Output
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 2
Enter length: 5
Enter width: 3
Area: 15.00
Perimeter: 16.00
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice:


Output
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 3
Enter first side: 3
Enter second side: 4
Enter third side: 5
Area: 6.00
Perimeter: 12.00
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice:


Output
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: 3
Enter first side: 1
Enter second side: 2
Enter third side: 5
Invalid triangle! The sum of any two sides must be greater than the third side.
===== SHAPE AREA & PERIMETER CALCULATOR =====
1. Circle
2. Rectangle
3. Triangle
4. Exit
Enter your choice: